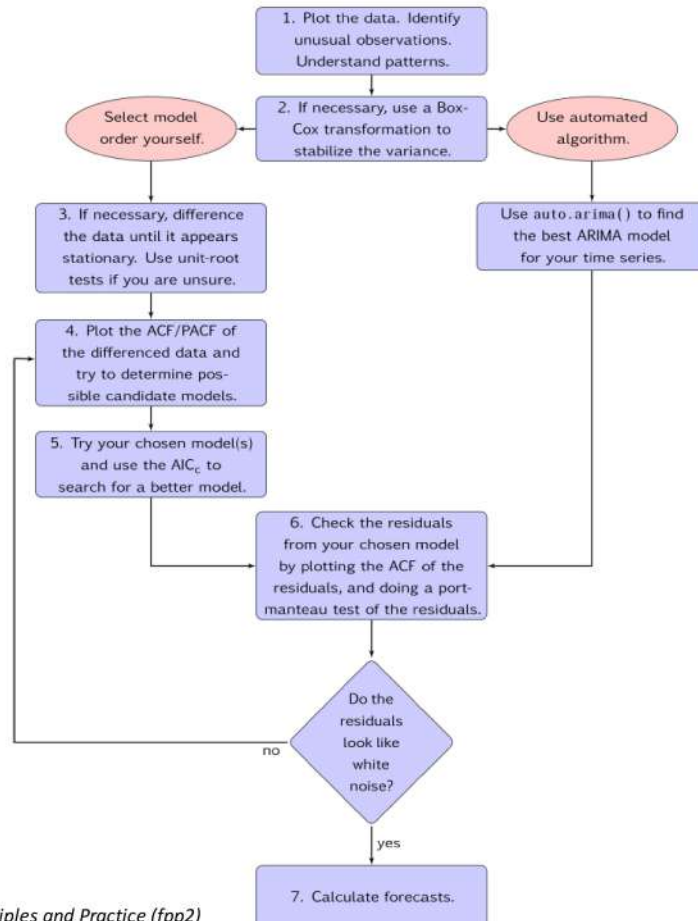




Taken from Forecasting: Principles and Practice (fpp2)

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Model choice: Using the ACF/PACF

Determining what ARIMA(p,d,q) model to fit based on the behaviour of the ACF and PACF

Given a stationary series that has significant autocorrelation at some lag g , how can I determine (a) What ARIMA(p,d,q) model is appropriate to fit to the data? and (b) What order (i.e. the values of p and q) model to fit?

ANSWER? Use the sample ACF and PACF plots. It is usually not possible to tell, simply from a time plot, what values of p and q are appropriate for the data. However, it is often possible to use the behaviour of the ACF and PACF plots. **ALWAYS LOOK AT THE COMBINED BEHAVIOUR OF THESE PLOTS.**

- If the time series is from an ARIMA(p,d,0) or ARIMA(0,d,q) process, then the ACF and PACF plots can help suggest the value of p or q. If p and q are both positive, then the plots do not help in finding suitable values of p and q.

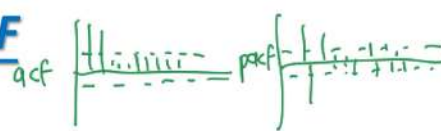
- The data may follow an ARIMA(p,d,0) model (i.e. an AR(p) model) if the ACF and PACF plots of the stationary (differenced) data show the following patterns:
 - i. the ACF is exponentially decaying or sinusoidal;
 - ii. there is a significant spike at lag p in the PACF, but none beyond lag p.

- The data may follow an ARIMA(0,d,q) model (i.e. an MA(q) model) if the ACF and PACF plots of the stationary (differenced) data show the following patterns:
 - i. the PACF is exponentially decaying or sinusoidal;
 - ii. there is a significant spike at lag q in the ACF, but none beyond lag q.

- If (a) BOTH the ACF and PACF decay slowly to 0 OR (b) BOTH the ACF and PACF have a sharp cut-off at low lag values then one would fit a low-order ARIMA(p,d,q) models and compare them with the AIC. Watch the relevant lecture video recording for examples of each of the above behaviours.

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Model choice: Using the ACF/PACF



Recognizing an ARIMA(p,d,0) [i.e. an AR(p)] model:

- Recall that partial autocorrelation is a measure of how much correlation between two variables at lag g is explained solely by their relationship, i.e. the partial autocorrelation function (abbreviated PACF) measures the excess correlation at lag g that has not already been accounted for by autocorrelations at lower lags.
- Thus, we can utilise the PACF plot to guide our choice of how many autoregressive terms to include in the model (when we observe slow decay in the ACF)

For example, the diagnostic patterns of the ACF and PACF of a AR(1) process:

PACF: cuts off abruptly after lag 1

ACF: declines/decays in a geometric/sinusoidal progression from its highest value at lag 1

- If we have significant partial autocorrelation coefficients up to lag 3, and slow decay to 0 in the ACF, then an AR(3) model may be appropriate.
- A PACF plot that displays no significant partial autocorrelations then an AR(p) model may not be appropriate for the data you are dealing with
- ***A further important result is that the values of the partial autocorrelation coefficients are the coefficients for the autoregressive terms you use in the AR(p) model***

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Model choice: Using the ACF/PACF

Recognizing an ARIMA(0,d,q) [i.e. MA(q)] model:

- We have already noted that the ACF of the MA(q) process has the distinctive feature of vanishing at lags greater than q . Thus we can utilise the sample ACF plot to guide our choice of how many moving average terms (q) to include in the model.
- This is because the observed value in time period t , X_t , is related to only the previous q random errors. Since the autocorrelation measures of the strength of the linear association (dependence) it makes sense that its value would be equal to 0 at lags greater than q .
- Essentially, the number of autocorrelation coefficients that are significant (outside the dashed lines) on the sample ACF plot indicate how many terms we should include in a MA(q) model (because those terms will be correlated with the observed value).

For example, the diagnostic patterns of the ACF and PACF of a MA(1) process:

ACF: cuts off abruptly after lag 1

PACF: declines/decays in a geometric/sinusoidal progression from its highest value at lag 1

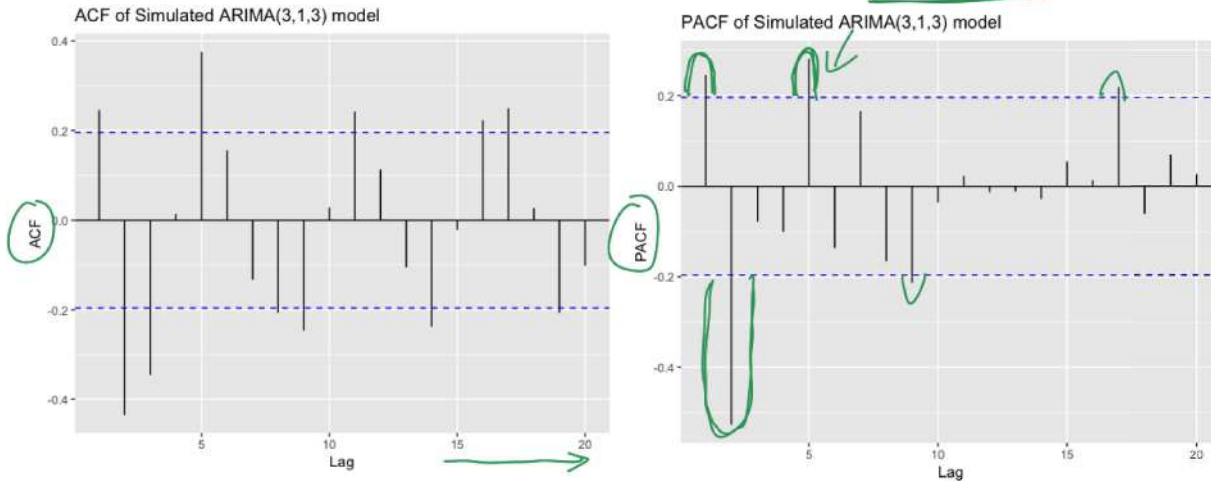
- If we have significant autocorrelation coefficients up to lag 3, then a MA(3) model may be appropriate.
- A sample ACF plot that displays no significant autocorrelations implies that a MA(q) model may not be appropriate for the data you are dealing with.

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Model choice: Using the ACF/PACF

Recognizing an ARIMA(p,d,q) [i.e. A mixed AR(p) - MA(q)] model:

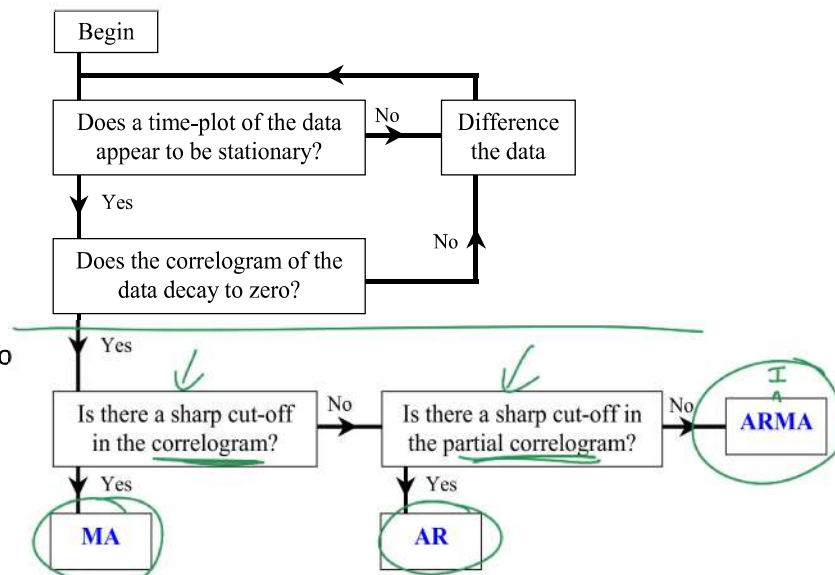
We have already noted that when there are both AR and MA terms in a model, then the behaviour of the ACF and PACF will not be helpful in determining the order of a model. At best, from their behaviour, we may be able to suggest that an ARIMA(p,d,q) model with $p > 0$ and $q > 0$ is needed. Consider the ACF and PACF plots below from a simulated ARIMA(3,1,3) model:



Neither the ACF nor the PACF shut off abruptly after a few lags, they both show slow decay to 0 (the ACF in a sinusoidal fashion, the PACF in a more geometric fashion). In a case like this, the best we can get from the behaviors shown here is that the must fit a mixed AR-MA model i.e. and ARIMA(p,d,q) model. We would start with low order models e.g. ARIMA(1,1,1), ARIMA(2,1,1), ARIMA(1,1,2), ARIMA(2,1,2) etc. and then compare them with AIC.

Model choice: Using the ACF/PACF

The flow diagram included here graphically displays the decision-making process involved when deciding on what type of ARIMA model to fit to a time series.

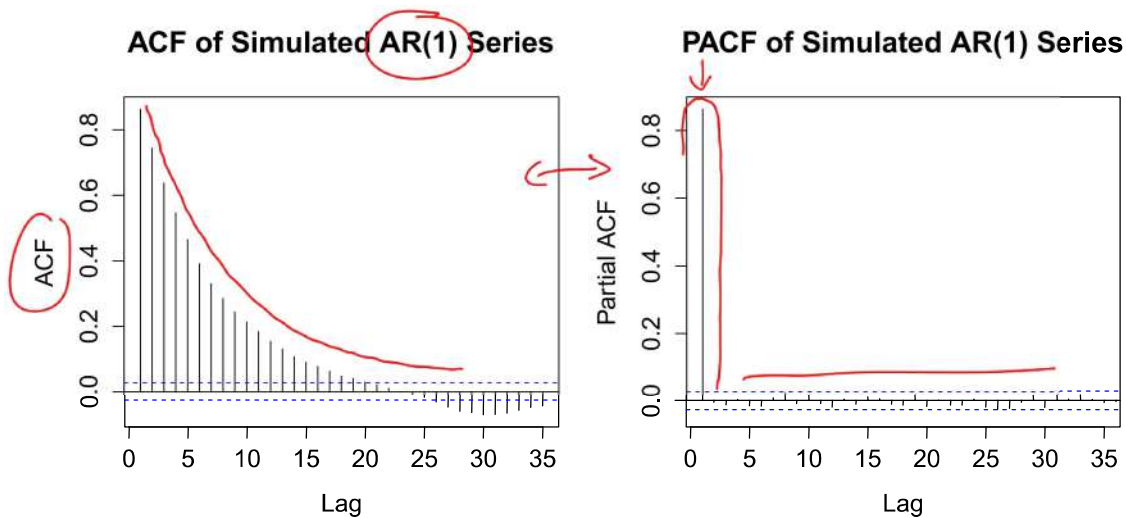


(From Diggle 2000, Fig. 6.2, page 169)

It is very important to understand that determining model order from ACF and PACF plots is **NOT** an exact science. They serve as a guide. Hence, we seldom only fit one ARIMA(p,d,q) model based on the behaviour of these plots. We always fit the suggested model and a few similar models and then compare them with AIC.

Model choice: Using the ACF/PACF

Consider the following example that displays the ACF and PACF of a simulated AR(1) time series for $T = 5000$, $\phi_1 = 0.86$:

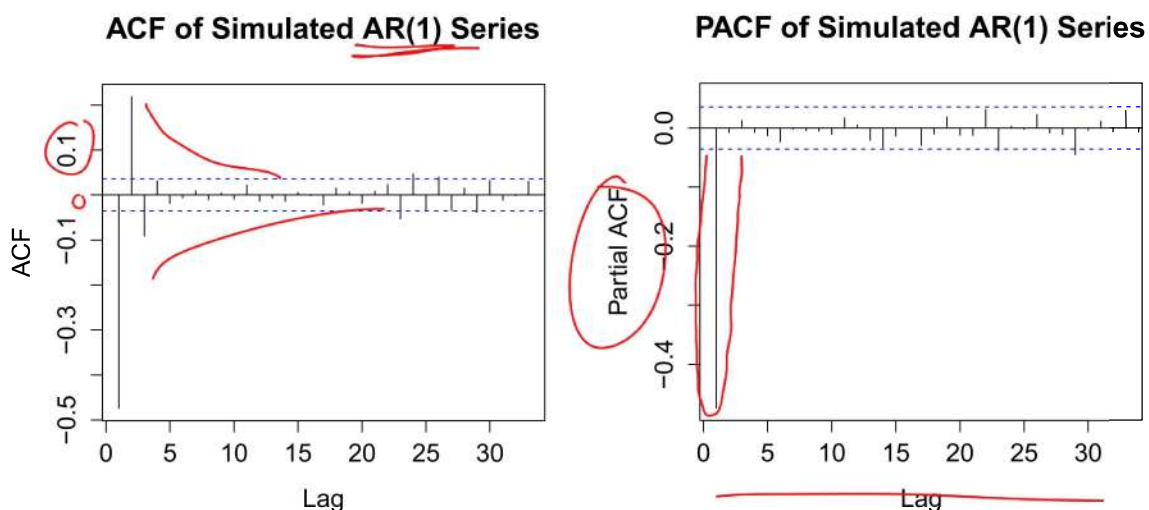


The ACF decays to 0 very slowly and in a sinusoidal fashion, but the PACF has a significant spike only at lag 1, after which it “shuts off” i.e. the partial autocorrelation function values at higher lags all lie between the dashed lines.

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Model choice: Using the ACF/PACF

Consider the following example that displays the ACF and PACF of a simulated AR(1) time series for $T = 3000$, $\phi_1 = -0.46$:

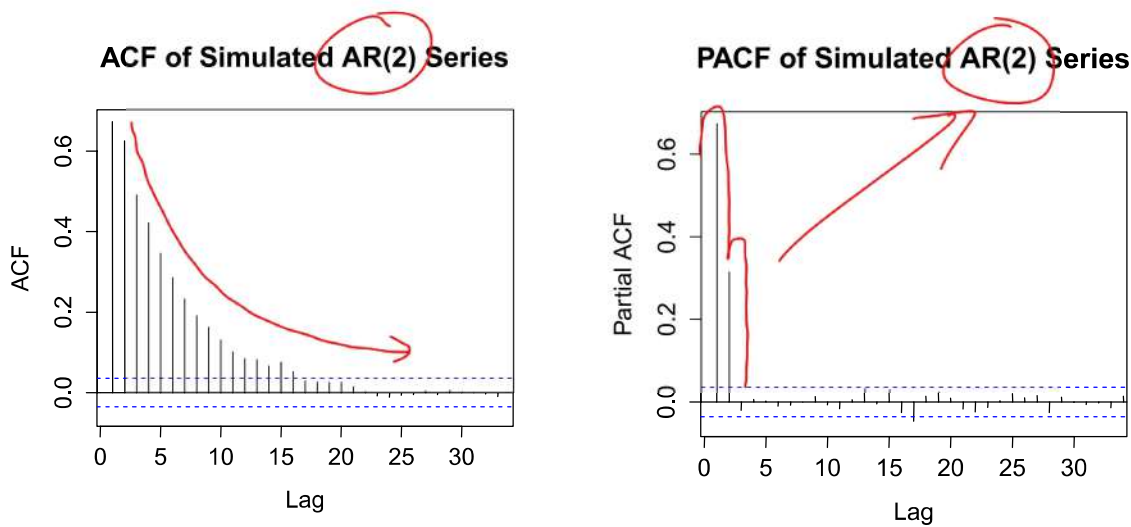


The ACF decays to 0 in a geometric fashion, but the PACF has a significant spike only at lag 1, after which it “shuts off” (the significant partial autocorrelation function value at lag 28 could be due to some form of sampling error).

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Model choice: Using the ACF/PACF

Consider the following example that displays the ACF and PACF of a simulated AR(2) time series for $T = 3000$, $\phi_1 = 0.66$ and $\phi_2 = 0.32$:

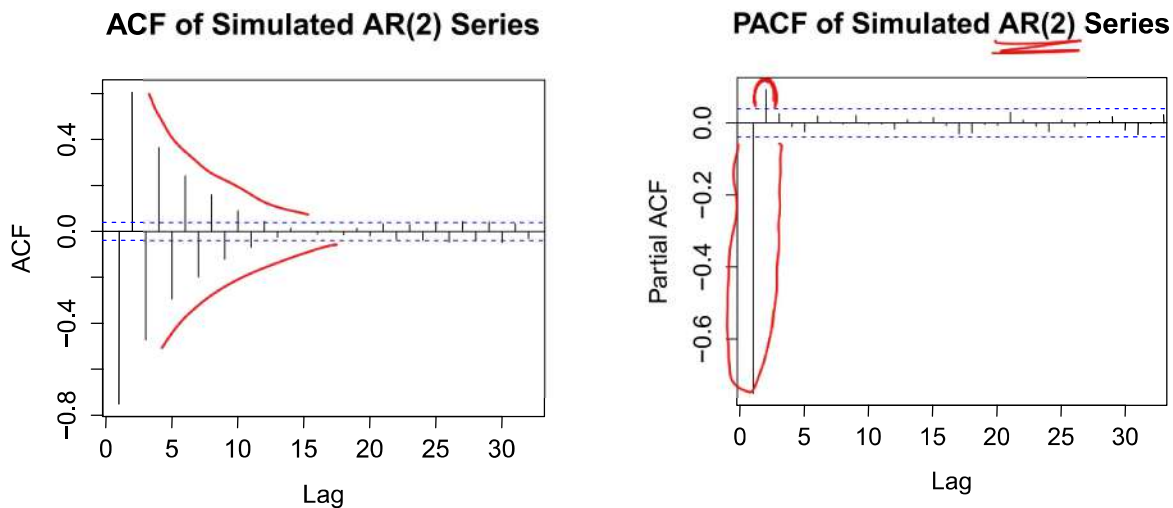


The ACF decays to 0 quite slowly, but the PACF has a significant spike only at lags 1 and 2, after which it “shuts off”

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Model choice: Using the ACF/PACF

Consider the following example that displays the ACF and PACF of a simulated AR(2) time series for $T = 3000$, $\phi_1 = -0.66$ and $\phi_2 = 0.12$:

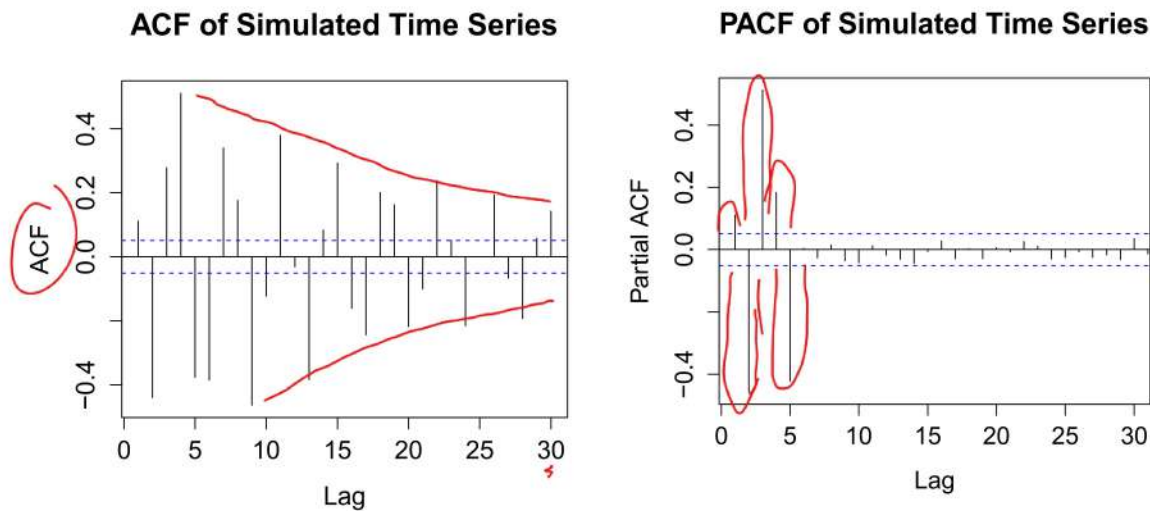


The ACF decays to 0 quite slowly, but the PACF has a significant spike only at lags 1 and 2, after which it “shuts off”

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Model choice: Using the ACF/PACF

Consider the following example of the ACF and PACF of a stationary series of data:

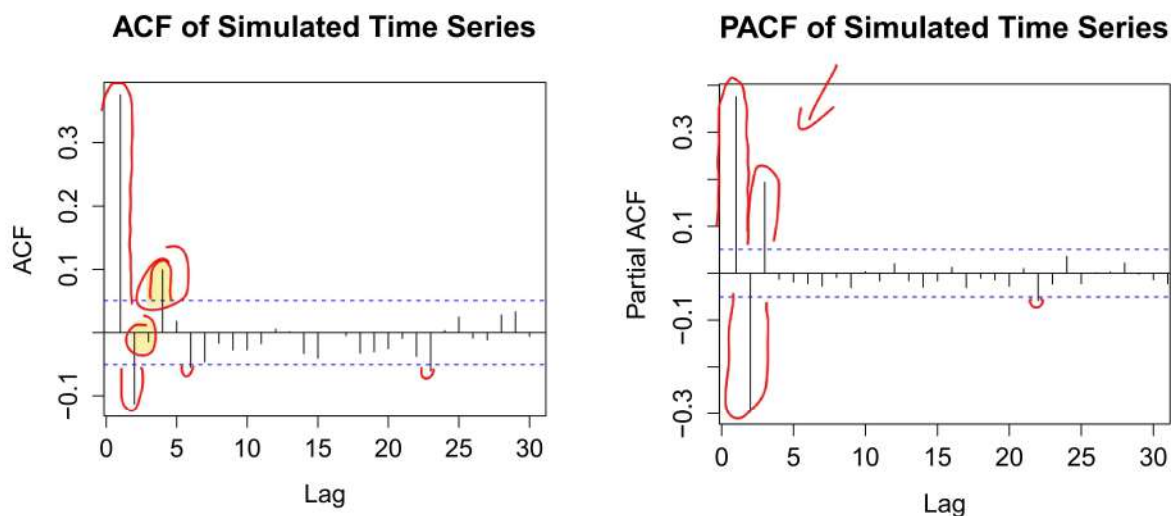


We see that the ACF slowly decays to 0, but the PACF shuts off abruptly after lag 5. Hence, an AR(5) model may be the appropriate order of AR model to fit to the data. So, we would fit an $ARIMA(5,d,0)$ model and then fit some other $ARIMA(p,d,q)$ models e.g. $ARIMA(2,1,2)$, $ARIMA(1,1,1)$ and then compare all models with AIC.

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Model choice: Using the ACF/PACF

Consider another example of the ACF and PACF of a stationary series of data:



We see that the ACF takes a while to decay to 0 as the lag increases, but the PACF shuts off abruptly after lag 3. Note the ACF at lags 6 & 23 is outside the 95% confidence bands. Its unlikely that both these are due to chance. There is also a potentially non-zero partial autocorrelation at lag 22, but as this is the only one just outside the bands at a higher lag value, we could say it is due to chance.

Handwritten notes: $ARIMA(3,d,0)$ $AR(3)$

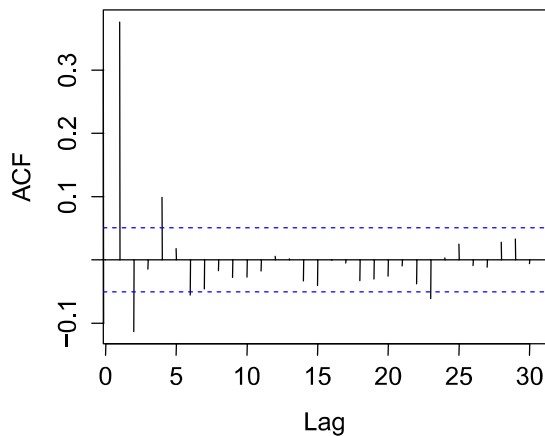
247

Q: If we chose to fit an $AR(p)$ model to this series, how many terms should be fit in the model?

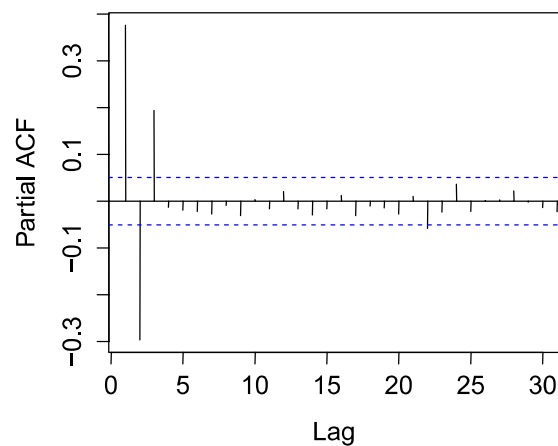
Model choice: Using the ACF/PACF

Consider another example of the ACF and PACF of a stationary series of data:

ACF of Simulated Time Series



PACF of Simulated Time Series



We see that the ACF looks like it decays to 0 as the lag increases, but the PACF shuts off abruptly after lag 3.

Q: If we chose to fit an $AR(p)$ model to this series, how many terms should be fit in the model? An $AR(3)$ model would be a good starting point.

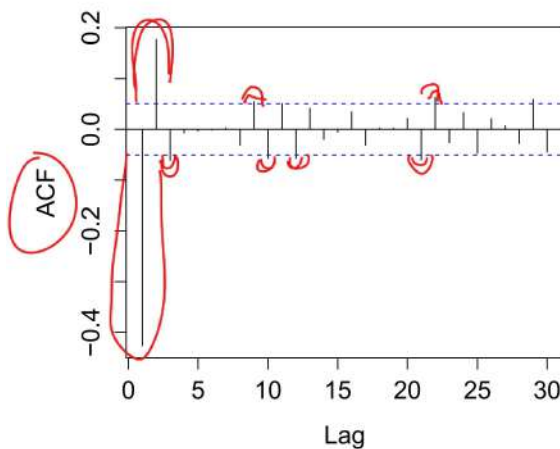
Again, we would fit an $ARIMA(3,d,0)$ model and then also some other $ARIMA(p,d,q)$ models and then compare them on their AIC values.

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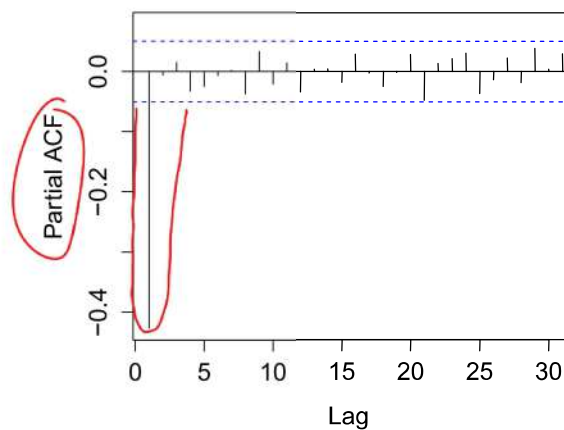
Model choice: Using the ACF/PACF

Consider the ACF and PACF plots of a stationary series below:

ACF of Simulated Time Series



PACF of Simulated Time Series

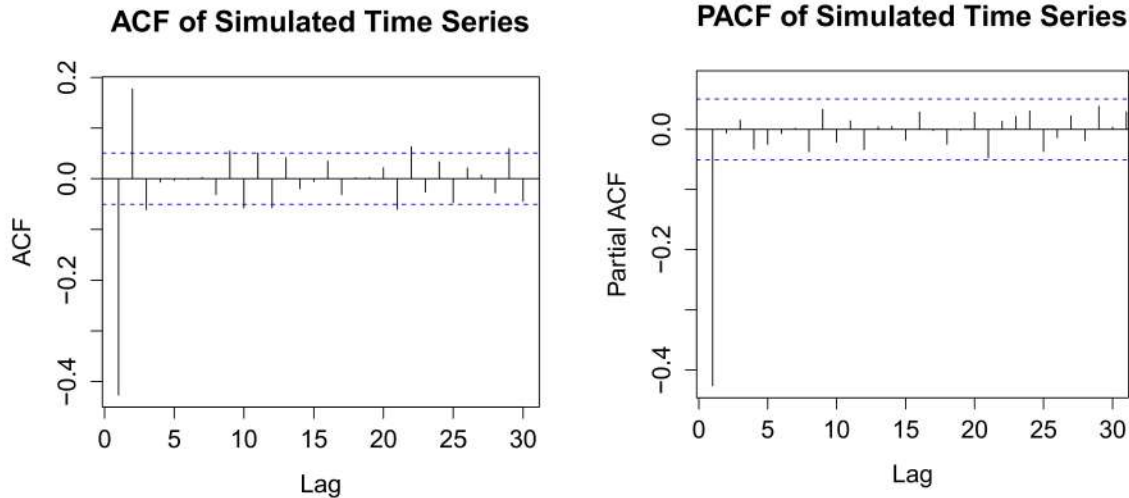


Q: What order of $AR(p)$ model should be fit here? $AR(1)$

249

Model choice: Using the ACF/PACF

Consider the ACF and PACF plots of a stationary series below:

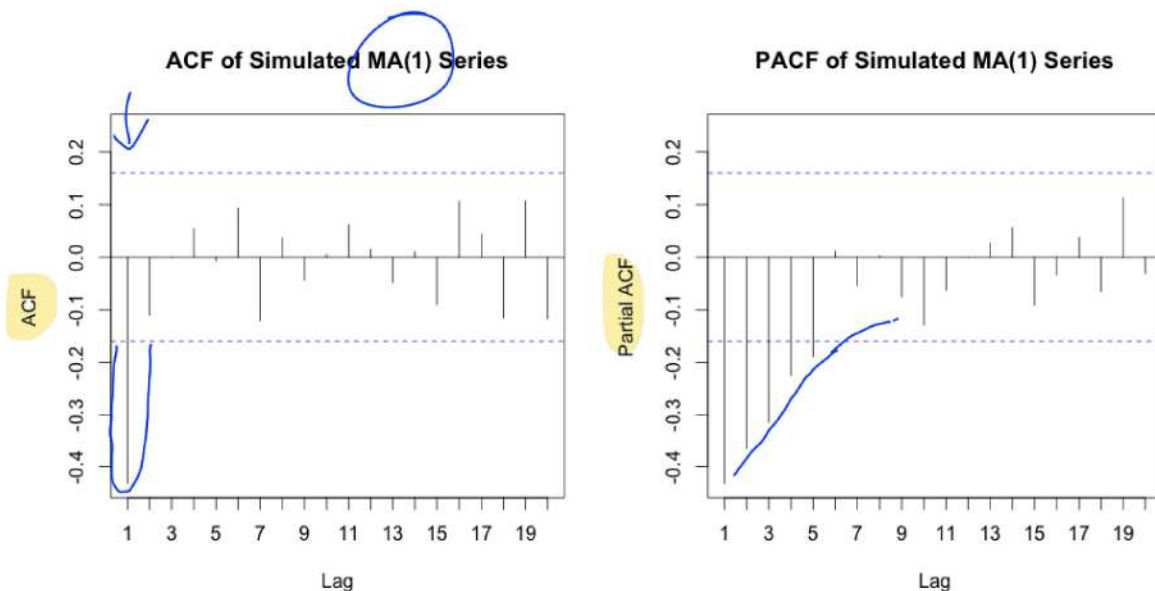


Q: What order of AR(p) model should be fit here? We see that the ACF decays slowly to 0 as the lag increases (there are several non-zero autocorrelations at higher lags in addition to the non-zero stronger autocorrelations at lags 1 and 2). The PACF shuts off abruptly after lag 1. Hence, an AR(1) model may be the appropriate order of AR model to fit to the data.

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Model choice: Using the ACF/PACF

Consider the ACF and PACF plots of a simulated MA(1) series with $\theta_1 = -1, n = 150$:

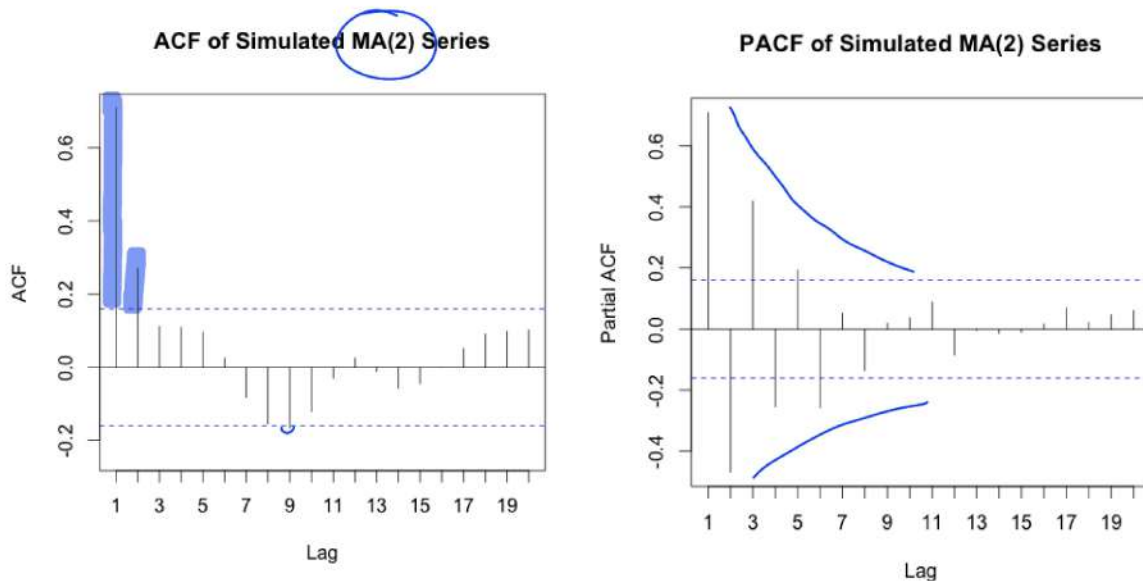


Note that the sample ACF has one significant spike at lag 1, after which it “shuts off”. The sample PACF decays slowly to 0 in a geometric fashion. Hence, a MA(1) model may be appropriate to fit to this series.

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Model choice: Using the ACF/PACF

Consider the ACF and PACF plots of a simulated MA(2) series with $\theta_1 = 2, \theta_2 = 1, n = 150$:

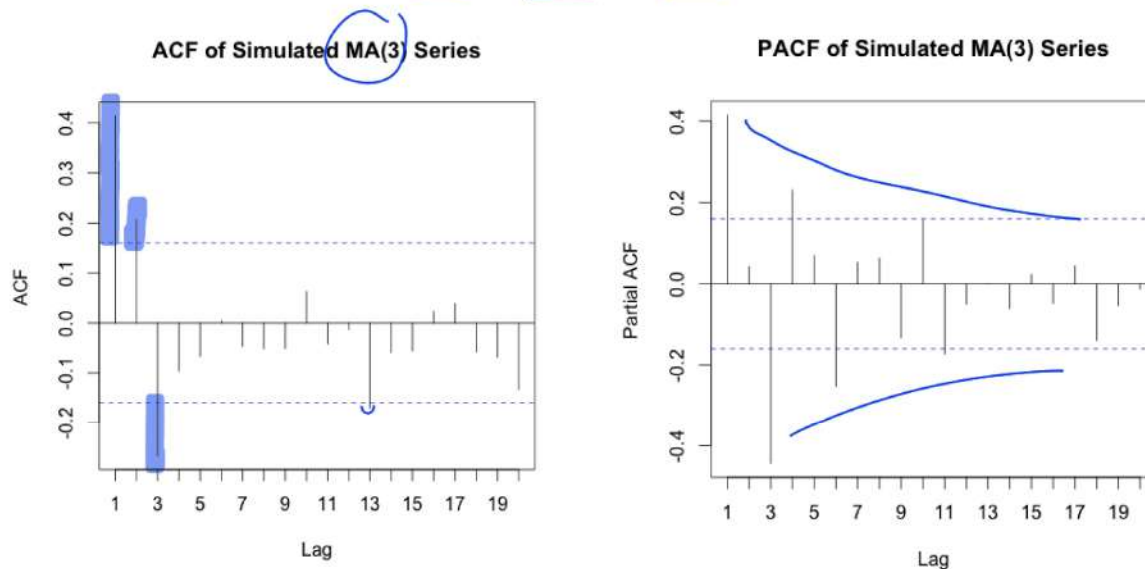


Note that the sample ACF is decaying sinusoidally to 0, with the strongest 2 autocorrelations at lags 1 and 2. Almost all the other autocorrelations are plausibly 0. The sample PACF decays slowly to 0 in a geometric fashion. Hence, a MA(2) model would be one of the ARIMA(p,d,q) models that we fit.

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Model choice: Using the ACF/PACF

Consider the ACF and PACF plots of a simulated MA(3) series with $\theta_1 = 0.8, \theta_2 = 0.7, \theta_3 = -0.6, n = 150$:

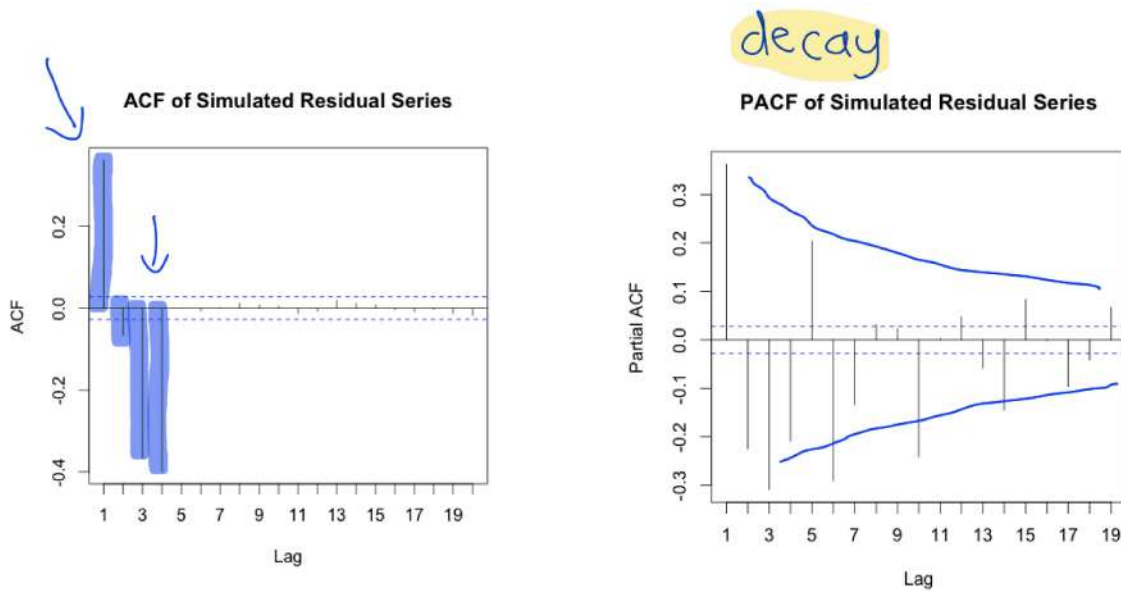


Note that the sample ACF has the three strongest autocorrelations at lags 1, 2 and 3. There is one non-zero autocorrelation at lag 13, which could be due to chance. The sample PACF decays slowly to 0 in a geometric fashion. Considering the combined behaviour of the plots, an MA(3) model is the suggested model to be fit. So we would fit this model as well as some other ARIMA(p,d,q) models and then compare them on AIC.

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Model choice: Using the ACF/PACF

Consider the ACF and PACF plots of a stationary series below:



We see that the ACF shuts off abruptly after lag 4, but the PACF slowly decays to 0.

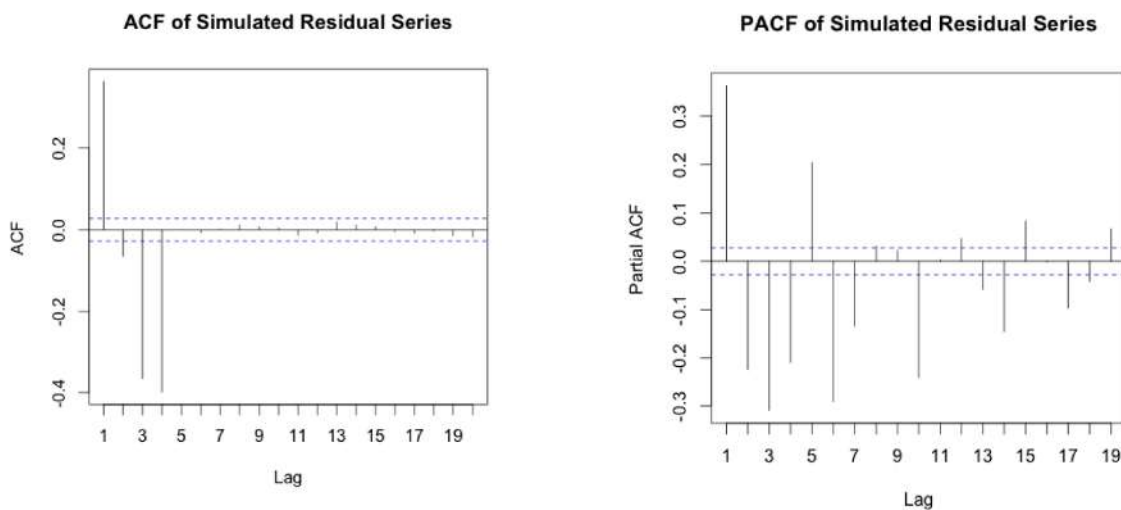
Q: Is it appropriate to fit a MA(q) model here? *MA(4)*

Q: If yes, what order of MA(q) model should be fit?

254

Model choice: Using the ACF/PACF

Consider the ACF and PACF plots of a stationary series below:



We see that the ACF shuts off abruptly after lag 4, but the PACF slowly decays to 0.

Q: Is it appropriate to fit a MA(q) model here? **YES**

Q: If yes, what order of MA(q) model should be fit? **A MA(4) model would be a good starting point. We would also fit a few other ARIMA(p,d,q) models.**

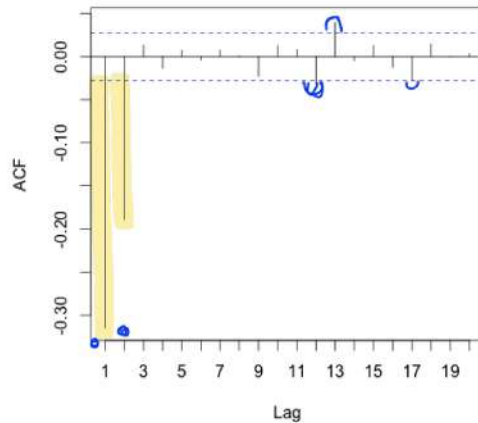
255

Model choice: Using the ACF/PACF

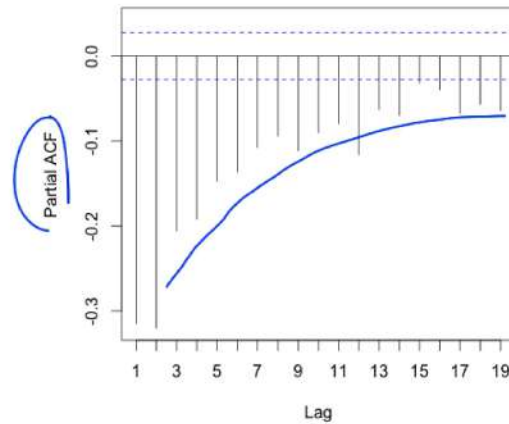
* MA $\rightarrow$ ACF (q)
 The man is a catfish
 * AR $\rightarrow$ PACF (p)
 Are you passing?

Consider the ACF and PACF plots of a stationary series below:

ACF of Simulated Residual Series



PACF of Simulated Residual Series



We see that the ACF has its strongest two autocorrelations at lags 1 and 2, with a few other non-zero (but very weak) autocorrelations at higher lag values. The PACF slowly decays to 0.

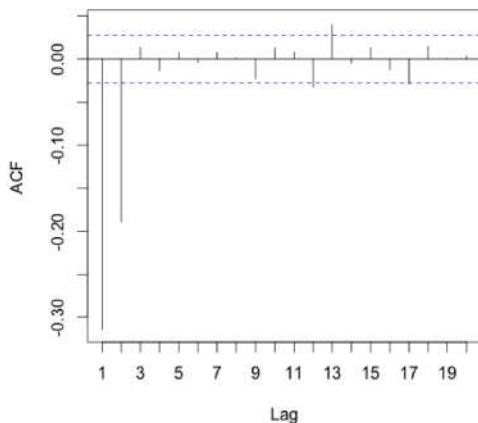
Q: What order MA(q) model should be fit here? MA(2)

256

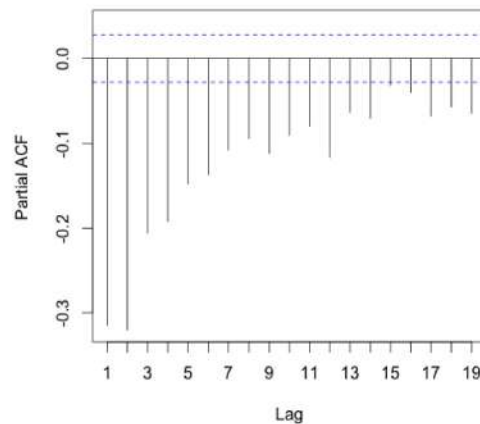
Model choice: Using the ACF/PACF

Consider the ACF and PACF plots of a stationary series below:

ACF of Simulated Residual Series



PACF of Simulated Residual Series



We see that the ACF has its strongest two autocorrelations at lags 1 and 2, with a few other non-zero (but very weak) autocorrelations at higher lag values. The PACF slowly decays to 0.

Q: What order MA(q) model should be fit here? **Considering the combined behaviour of the two plots, a MA(2) model would be appropriate to fit to this time series.**

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